

Experiment 8

1).Develop a program using malloc(), calloc(), realloc(), and free(). Monitor memory allocation behavior and identify memory leaks using Valgrind. **Code:**

```
#include <stdio.h>
```

```
#include <stdlib.h>
```

```
int main() {
```

```
    int i;
```

```
    // 1. malloc()    int *a = (int  
    *)malloc(3 * sizeof(int));
```

```
    if (a == NULL) {  
printf("malloc failed\n");  
return 1;  
    }
```

```
    printf("Memory allocated using malloc()\n");
```

```
    for (i = 0; i < 3; i++) {        a[i] =  
(i + 1) * 10;        printf("a[%d] =  
%d\n", i, a[i]);  
    }
```

```
    // 2. calloc()    int *b = (int  
    *)calloc(3, sizeof(int));    if (b ==
```

```
NULL) {    printf("calloc failed\n");  
free(a);    return 1;  
}
```

```
printf("\nMemory allocated using calloc()\n");
```

```
for (i = 0; i < 3; i++) {  
printf("b[%d] = %d\n", i, b[i]);  
}
```

```
// 3. realloc()    a = (int  
*)realloc(a, 5 * sizeof(int));
```

```
if (a == NULL) {  
printf("realloc failed\n");  
free(b);    return 1;  
}
```

```
printf("\nMemory resized using realloc()\n");
```

```
a[3] = 40;  
a[4] = 50;    for  
(i = 0; i < 5; i++)  
{  
printf("a[%d] =  
%d\n", i, a[i]);
```

```
}
```

```
// 4. free()
```

```
free(a);
```

```
printf("\nMemory allocated by malloc/realloc has been freed.\n");
```

```
free(b); printf("Memory allocated by calloc has been  
freed.\n");
```

```
return 0;
```

```
}
```

```
venkat038@AYYAPPA:~$ nano 8.1.c  
venkat038@AYYAPPA:~$ gcc 8.1.c -o 8.1  
venkat038@AYYAPPA:~$ ./8.1  
Memory allocated using malloc()  
a[0] = 10  
a[1] = 20  
a[2] = 30  
  
Memory allocated using calloc()  
b[0] = 0  
b[1] = 0  
b[2] = 0  
  
Memory resized using realloc()  
a[0] = 10  
a[1] = 20  
a[2] = 30  
a[3] = 40  
a[4] = 50  
  
Memory allocated by malloc/realloc has been freed.  
Memory allocated by calloc has been freed.  
venkat038@AYYAPPA:~$ |
```

2).Develop a program that demonstrates Copy-on-Write behavior after a fork(). Analyze memory usage before and after modifying data in the child process.

```
#include <stdio.h>
```

```
#include <stdlib.h>
```

```
#include <unistd.h>
```

```
#include <sys/wait.h>
```

```
#define SIZE (10 * 1024 * 1024) // 10 MB
```

```
int main() {
```

```
    char *data = malloc(SIZE);
```

```
    if (data == NULL) {        printf("Memory  
allocation failed\n");    return 1;
```

```
}
```

```
    // Fill memory so that it is actually  
    allocated    for (int i = 0; i < SIZE; i++) {  
data[i] = 'A';  
    }
```

```
    printf("Parent: 10 MB memory allocated.\n");
```

```
    printf("Parent PID: %d\n", getpid());
```

```
pid_t pid = fork();

if (pid < 0) {
printf("fork() failed\n");
free(data);    return 1;
}

if (pid == 0) {    // Child process    printf("\nChild PID: %d\n",
getpid());    printf("Child: Memory is initially shared using Copy-on-
Write.\n");

sleep(2);

// Modify the memory    printf("Child:
Modifying 10 MB of memory...\n");

for (int i = 0; i < SIZE; i++) {
data[i] = 'B';
}

printf("Child: Memory modification completed.\n");

sleep(5);
```

```
        free(data);
    exit(0);
}
else {
    // Parent process
    printf("\nParent: Child created using fork().\n");    printf("Parent: Both
processes initially share the same memory pages.\n");

    sleep(8);

    wait(NULL);

    printf("\nParent: Child process completed.\n");

    free(data);
}

return 0;
}
```

```
venkat038@AYYAPPA:~$ nano 8.2.c
venkat038@AYYAPPA:~$ gcc 8.2.c -o 8.2
venkat038@AYYAPPA:~$ ./8.2
Parent: 10 MB memory allocated.
Parent PID: 751

Parent: Child created using fork().
Parent: Both processes initially share the same memory pages.

Child PID: 752
Child: Memory is initially shared using Copy-on-Write.
Child: Modifying 10 MB of memory...
Child: Memory modification completed.

Parent: Child process completed.
venkat038@AYYAPPA:~$ |
```